

Subset Sum

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Set of n numbers - $\{a_1, a_2, a_3, \dots, a_n\}$

subset of numbers whose sum is " w ".

~~Ex: $\{6, 2, 3, 1\}$~~

[Ex:] $\{6, 2, 3, 1\}$ - set of 4 numbers.

is there ^{any} subset of this 4 numbers whose sum is '5'.

$\rightarrow \{2, 3\} \rightarrow 2+3 \rightarrow 5$

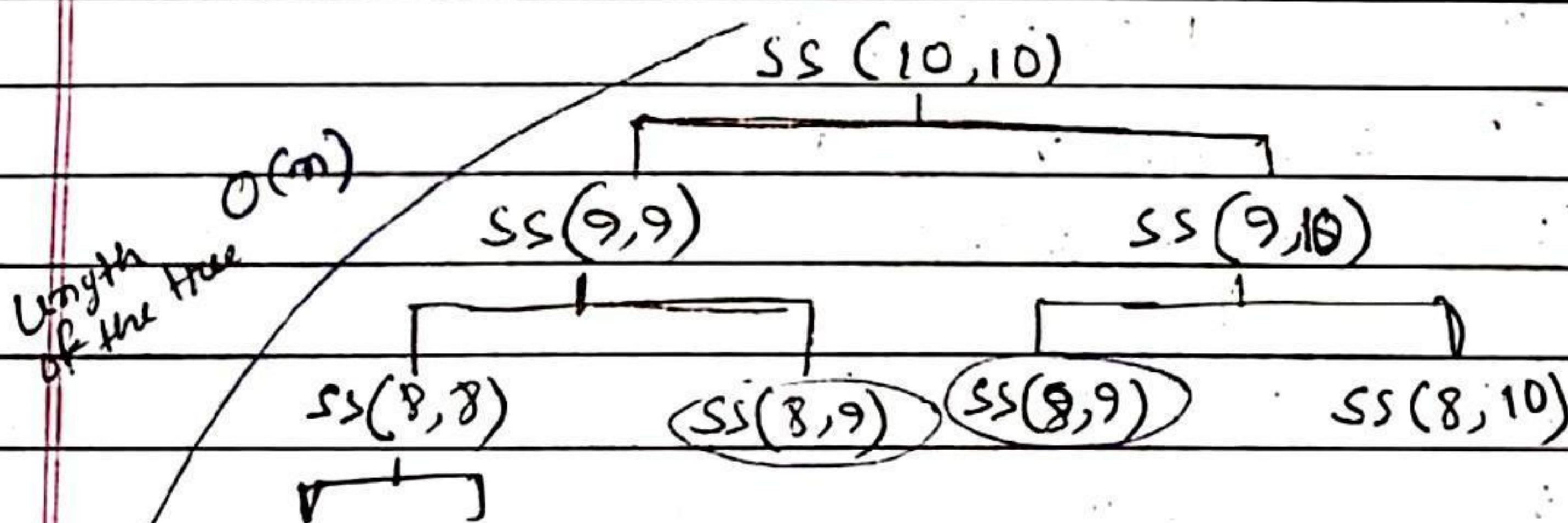
Recursive equation =

$$SS(i, s) = \begin{cases} SS(i-1, s) & ; s < a_i \\ SS(i-1, s-a_i) \vee SS(i-1, s) & ; s \geq a_i \\ \text{True} & ; s = 0 \\ \text{False} & ; i = 0, s \neq 0 \end{cases}$$

$\{1, 2, 3, 4\}$

10-4

Recursive tree :



no of nodes here - $O(2^n)$

Time complexity = $O(2^n)$

$SS(n, w)$, unique subproblem $O(nw)$.

Example-①

$$SS(i, s) = \begin{cases} SS(i-1, s) & ; s < a_i \\ SS(i-1, s-a_i) \vee SS(i-1, s) & ; s \geq a_i \\ \text{True} & ; s=0 \\ \text{False} & ; i=0, s \neq 0 \end{cases}$$

^{1 2 3 4}
 $\{6, 3, 2, 1\}$ - 4 element

$W = 5$ (sum)

^{i = 1 2 3 4}

6	3	2	1
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using Bottom up dynamic programming -

		0	1	2	3	4	5
i = 0		T	F	F	F	F	F
(1 ele) i = 1		T	F	F	F	F	F
2 (element) i = 2		T	F	F	T	F	F
3 (ele) i = 3		T	F	T	T	F	T
4 (ele) i = 4		T	T	T	T	T	T

Time complexity = $O(mw)$

Space complexity = $O(mw)$

Final answer

$$SS(1, 1) = SS(0, 1)$$

(the sum of subset = 5)

→ True

$$SS(2, 1) = SS(1, 1)$$

$$SS(2, 3) = SS(1, 0) \vee SS(1, 3)$$

$$SS(2, 4) = SS(1, 1) \vee SS(1, 4)$$

$$SS(2, 5) = SS(1, 2) \vee SS(1, 5)$$

$$SS(3, 1) = SS(2, 1)$$

$$SS(3, 2) = SS(2, 0) \vee SS(2, 2)$$

$$SS(3, 3) = SS(2, 1) \vee SS(2, 3)$$

$$SS(3, 4) = SS(2, 2) \vee SS(2, 4)$$

$$SS(3, 5) = SS(2, 3) \vee SS(2, 5)$$

$$SS(4, 1) = SS(3, 0) \vee SS(3, 1)$$

$$SS(4, 2) = SS(3, 1) \vee SS(3, 2)$$

$$SS(4, 3) = SS(3, 2) \vee SS(3, 3)$$

$$SS(4, 4) = SS(3, 3) \vee SS(3, 4)$$

$$SS(4, 5) = SS(3, 4) \vee SS(3, 5)$$